روبوتات الشرق الأوسط وشمال أفريقيا Robotics MENA

RESEARCH REPORT

ENERGY · ADNOC · ADIPEC 2026

ADNOC's real AI story at ADIPEC wasn't the OpenAI deal. It was the robots.

Wall-climbing crawlers, explosion-proof inspectors and remotely operated ships: after a year of war damage, the Gulf's largest energy company is buying robots as infrastructure. It should now publish the results.

Robotics MENA analysis 7 Nov 2026



Gecko's Raptor at ADIPEC, Abu Dhabi, November 2026

Photo: Victor Besa / The National

70
Robotics use cases deployed or in development, per ADNOC

5
Robotics providers profiled in this report

AED110m
Ceiling of the 2024 ADNOC Gas–Gecko contract

60%
Of Habshan's processing capacity restored after the strikes

roboticsmena.com · contact@roboticsmena.com

Read the interactive version: roboticsmena.com/spotlight?s=gecko-adnoc-raptor

CONTENTS

01 The overlooked story
02 Why inspection, why now
03 Not a pilot programme
04 The numbers deserve scrutiny
05 Our view
06 What we're watching
07 About the providers
— Sources

KEY FINDINGS

01
The real story is physical
The OpenAI deal took the headlines, but robots on steel walls, plant floors and ships will shape the Gulf market more.

02
War made inspection a resilience problem
After strikes on Shah, Habshan and Ruwais, every damaged metre of steel must be checked before it carries pressure again.

03
ADNOC buys at scale, from many suppliers
Crawlers, explosion-proof inspectors, an automated rig, remotely operated ships and humanoid trials are all part of the push.

04
The numbers still need proof
Most headline figures are company claims, projections or contract ceilings. ADNOC should publish measured results.

HOW TO READ THIS REPORT

Every figure is sourced. Headline claims are rated by the strength of the evidence behind them:

- Company claim
- Single source or projection
- Documented, with caveats
- Reported by several outlets

01 CHAPTER 01

The overlooked story

When ADNOC signed with OpenAI and G42 at ADIPEC on 2 November, the headlines wrote themselves: the first energy company to deploy OpenAI's Frontier agent platform. But Frontier is software for planners and traders. It does not climb, crawl or turn a valve. The more consequential story of the week was on the exhibition floor, clinging to a steel wall.

There, Gecko Robotics unveiled Raptor, a wall-climbing inspection robot that ADNOC is deploying across its facilities to find structural weaknesses and assess damage. Gecko says it is three to five times faster than its previous robot. On 5 November ADNOC Onshore signed a new agreement with Gecko to extend robotic inspection across its field operations.

Announced 2 Nov 2026

OpenAI Frontier

Decision AI for planners and traders

- Agreement between ADNOC, OpenAI and G42
- Software agents for field planning and valuing each barrel
- No robots, drones or field automation

Unveiled 4 Nov 2026

Gecko Raptor

Robots on steel

- Wall-climbing inspection robot shown at ADIPEC
- Finds structural weaknesses and assesses damage
- Being deployed across ADNOC facilities

02 CHAPTER 02

Why inspection, why now

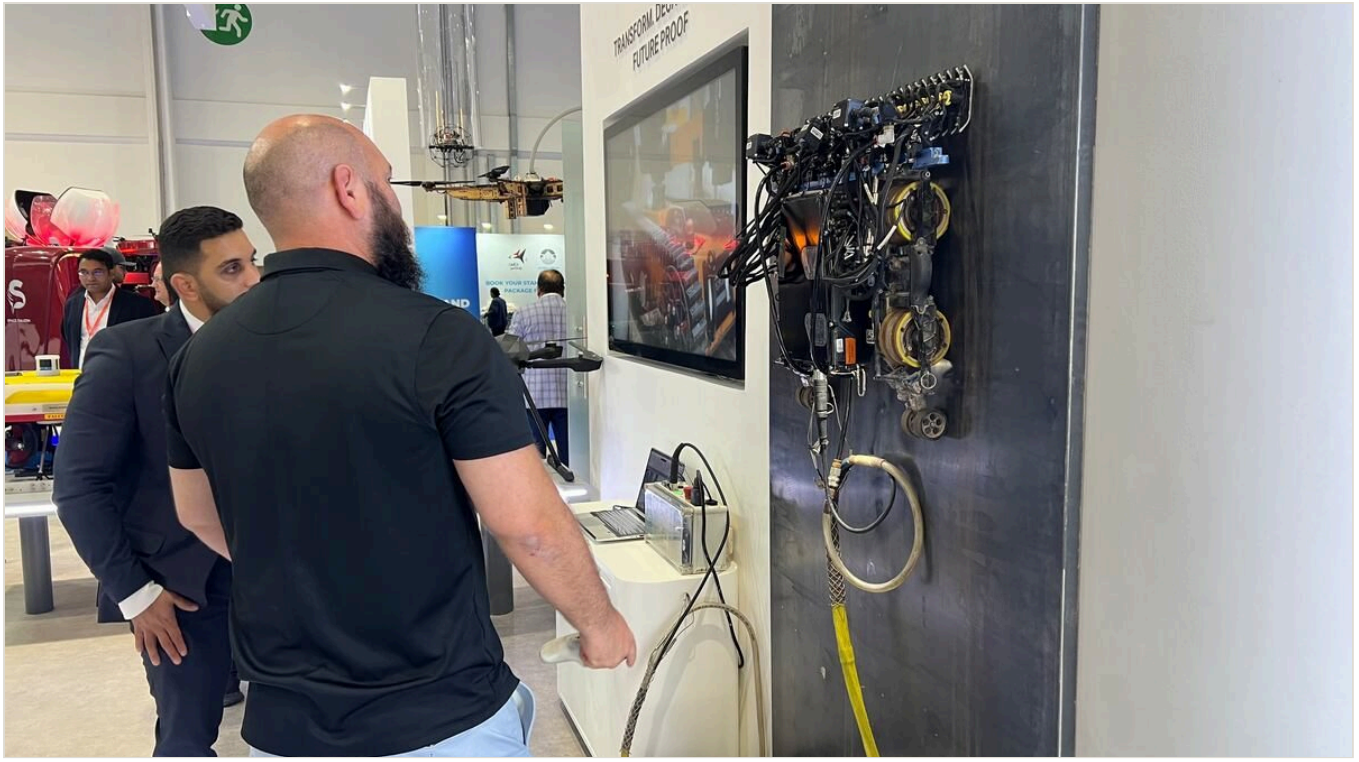
The timing is not abstract. In March and April, Iranian missiles and drones struck ADNOC assets. A drone attack set fire to the Shah gas field on 16 March and suspended operations. Habshan was hit repeatedly, and on 3 April falling debris killed a contractor's engineer and injured four others. Fires broke out at the Ruwais refinery. ADNOC Gas says it restored 60% of Habshan's processing capacity soon after and is aiming for 80% by the end of 2026.

ATTACK AND RECOVERY LOG

16 Mar 2026	Shah gas field: drone attack causes a fire; operations suspended Source: gCaptain / Bloomberg
3 Apr 2026	Habshan hit again; falling debris kills a contractor's engineer and injures four Source: gCaptain / Bloomberg
5 Apr 2026	Fires reported at the Ruwais refinery Source: gCaptain / Bloomberg
12 May 2026	ADNOC Gas: 60% of Habshan capacity restored, 80% targeted by end-2026 Source: Khaleej Times



Every metre of damaged steel has to be checked before it carries pressure again. Done the old way, that means scaffolding, rope access, shutdowns and inspectors who are in short supply. Gecko's crawlers read wall thickness with sensors and sound waves while the plant keeps running. "Robotics are essential to ensure that resilience and readiness is possible in a world where we don't have a lot of these skilled workforces," Gecko's chief executive, Jake Loosararian, told The National.



A Gecko crawler detects corrosion in places that are hard to reach by hand. Photo: Cody Combs / The National

03 CHAPTER 03

Not a pilot programme

What sets ADNOC apart is how quickly it commits at scale. ADNOC Gas first used Gecko's robots in 2023. In April 2024 it signed a contract worth up to AED110 million with Gecko and Al Masaood Energy; Gecko reported that the pilot raised asset coverage by 99.6% and projected a one-third cut in lost-time injuries from removing scaffolding. In November 2025 came a three-year programme with AIQ that ADNOC Gas says could save more than $300 million in maintenance and inspection over five years. Gecko now says it deploys tens of robots a day in the region.

> **"There aren't a lot of proof-of-value meetings; there's very much a 'let's scale it' type of thinking."**
>
> Jake Loosararian, Gecko Robotics, on working with ADNOC (The National, August 2026)

ADNOC ROBOTICS TIMELINE

2023

ADNOC Gas first uses Gecko robots

ADNOC Gas becomes Gecko's first client in the UAE, using its wall-climbing robots on gas assets.

Source: ADNOC Gas

29 Apr 2024

AED110m contract with Gecko and Al Masaood Energy

A contract worth up to AED110 million. Gecko says the pilot raised asset coverage by 99.6% and that removing scaffolding could cut lost-time injuries by a third.

Source: Gecko Robotics

2 Nov 2025

Three-year AIQ–Gecko–ADNOC Gas programme

A programme to digitise the full inspection-to-decision workflow. ADNOC Gas projects more than $300 million in savings over five years.

Source: ADNOC Gas

13 Nov 2025

ADNOC L&S unveils remotely operated landing craft

Two 60-metre vessels, built with France's SeaOwl and steered from a control centre in Mussafah.

Source: Zawya (ADNOC L&S release)

21 May 2026

Taurob inspector goes live at Taweelah

An explosion-proof robot starts autonomous inspections for gas leaks and hotspots at the Taweelah gas compression plant.

Source: Hydrocarbon Processing

25 Jun 2026

AD-300 island rig delivered early

ADNOC Drilling's first island rig arrives nearly three months ahead of schedule, with automated pipe handling.

Source: OE Digital

4 Nov 2026

Raptor unveiled at ADIPEC

Gecko shows its new wall-climbing robot, which ADNOC is deploying to find structural weaknesses and assess damage.

Source: The National

4 Nov 2026

ADNOC Onshore to test Unitree robots

An agreement to test Unitree robots, including humanoids, in field operations, reported by Attaqa.

Source: Attaqa (الطاقة)

5 Nov 2026

New ADNOC Onshore–Gecko agreement

Signed at ADIPEC to extend robotic inspection across ADNOC Onshore's field operations.

Source: ADNOC Group on X

Dec 2026

ARGOS operator robot due

The heavy-duty operator robot, designed to turn valves and read gauges, is due in service by the end of 2026.

Source: Hydrocarbon Processing

Gecko is only one supplier. At the Taweelah gas compression plant, an explosion-proof Taurob robot has been running autonomous inspections for gas leaks and hotspots since May, using LiDAR and thermal cameras. A heavier "operator" robot, ARGOS, designed to turn valves and read gauges, is due in service by the end of 2026. ADNOC Drilling's AD-300 island rig, delivered in June nearly three months early, uses automated pipe handling. ADNOC L&S has ordered two 60-metre remotely operated landing craft, steered from a control centre in Mussafah. And according to the energy news site Attaqa, ADNOC Onshore has agreed to test Unitree robots, including humanoids, in its field operations.

ROBOT FLEET

● Deploying

Raptor

Wall-climbing crawler · Gecko Robotics
Where: ADNOC facilities
Finds structural weaknesses and assesses damage, scanning steel with sensors and sound waves without taking assets offline.

3–5× faster than Gecko's previous robot (company claim)

Source: The National

● Live

Taurob inspector

Explosion-proof ground robot · Taurob
Where: Taweelah gas compression plant
Runs autonomous inspections for gas leaks and hotspots, using LiDAR and thermal cameras.

In live operations since May 2026

Source: Hydrocarbon Processing

● Due

ARGOS

Heavy-duty operator robot · Taurob (joint industry project)
Where: Taweelah
Designed to lift, grip, turn valves and operate gauges. Partners include Equinor, TotalEnergies and Petrobras.

Due in service by end-2026

Source: Hydrocarbon Processing

● Delivered

AD-300 island rig

Automated drilling rig · ADNOC Drilling
Where: Artificial island, offshore Abu Dhabi
The first of six island rigs, with automated pipe handling and AI-enabled monitoring.

Delivered June 2026, nearly three months early

Source: OE Digital

Ordered

Remote landing craft

Remotely operated vessel · ADNOC L&S with SeaOwl
Where: Controlled from Mussafah
Two 60-metre vessels with no crew accommodation, steered from a remote control centre.

Delivery: late 2026 or mid-2027 (sources differ)

Source: Zawya (ADNOC L&S release)

Trial agreed

Unitree robots

Quadrupeds and humanoids · Unitree Robotics
Where: ADNOC Onshore field operations
An agreement to test Unitree robotics and humanoid technologies in ADNOC Onshore's field operations.

Single source: Attaqa

Source: Attaqa (الطاقة)



ADNOC's Taurob inspector, deployed at the Taweelah gas compression plant in May 2026. Image: ADNOC, via JPT

AED110m
Ceiling of ADNOC Gas's 2024 Gecko contract

$300m+
Projected five-year savings, AIQ–Gecko programme (company claim)

70
Robotics use cases "deployed or under development" (ADNOC)



Rendering of the 60-metre remotely operated landing craft ordered by ADNOC L&S. Image: ADNOC L&S

04 CHAPTER 04

The numbers deserve scrutiny

Not all of this is as solid as the press releases suggest. Tayba Al Hashmi, chief executive of ADNOC Offshore, told an ADIPEC panel that the group has "70 robotics use cases deployed or under development". That figure blends working robots with projects on paper, and no public breakdown exists. Raptor's headline claims, up to 50 times faster than competitors' robots and a 50-fold cut in data-collection cost, come from Gecko and have not been independently tested. Delivery dates drift too: ADNOC L&S said in November 2025 that its landing craft would arrive in late 2026, while its fleet team later pointed to mid-2027.

Two figures are easy to confuse. Gecko's $300 million US Navy deal is a contract ceiling, not guaranteed spending. ADNOC Gas's $300 million is a projected saving. Neither is a measured result.

CLAIM CHECK

"70 robotics use cases"

ADNOC Offshore CEO, ADIPEC panel

No breakdown

WHAT WE FOUND

Reported once, by Khaleej Times. The figure counts projects deployed and under development together, and ADNOC has published no list.

Source: Khaleej Times

Raptor "up to 50× faster than competitors"

Gecko Robotics, via The National

Company claim

WHAT WE FOUND

Not independently tested. We found no other outlet reporting Raptor's performance.

Source: The National

"50× lower data-collection cost"

Gecko Robotics, via The National

Company claim

WHAT WE FOUND

Rests on the chief executive's statement; no method or baseline was published.

Source: The National

"$300m+ savings over five years"

ADNOC Gas press release

Projection

WHAT WE FOUND

A forecast for the AIQ–Gecko programme, not money saved so far.

Source: ADNOC Gas

"$300m US Navy deal"

Gecko Robotics

Contract ceiling

WHAT WE FOUND

A real contract, but $300m is the ceiling of an IDIQ award, not guaranteed spending. It is unrelated to ADNOC.

Source: Gecko Robotics

Landing craft "delivered in Q4 2026"

ADNOC L&S, November 2025

Conflicting dates

WHAT WE FOUND

ADNOC L&S's fleet team later told Riviera the two vessels are contracted for Q2–Q3 2027.

Source: Riviera

Taurob robot "in live operations" at Taweelah

ADNOC release, reported by three outlets

Widely reported

WHAT WE FOUND

Carried by Hydrocarbon Processing, Interesting Engineering and JPT, with specifications from Taurob.

Source: Hydrocarbon Processing

05 CHAPTER 05

Our view

> **ADNOC is doing what most operators in the region have not: buying robots as infrastructure rather than as demonstrations. The war turned inspection from a cost line into a question of resilience, and ADNOC moved accordingly. That is the right call. It will shape the Gulf's robotics market more than any agent platform, because procurement at this scale builds the service crews, the data and possibly the local manufacturing that others can use.**

Gecko's chief executive calls ADNOC "a technology company that happens to make oil and gas". If so, it should report like one. Publish what the 70 use cases are and which are live. Report measured outcomes, such as inspection hours, shutdown days avoided and injuries, rather than projected savings. And say whether the UAE robot manufacturing discussed since 2025 is actually happening.

06 CHAPTER 06

What we're watching

WATCHLIST (AS OF 11 NOV 2026)

Due	ARGOS, the heavy-duty operator robot, entering service at Taweelah by the end of 2026.	By end-2026 In 7 weeks
Open question	How many Raptors ADNOC fields, and the first published damage-assessment results.	Ongoing
Trial	ADNOC Onshore's trials of Unitree robots, including humanoids.	Agreed Nov 2026
Due	Delivery of the remotely operated landing craft: late 2026 or 2027.	Late 2026 or 2027
Open question	Whether Gecko starts building robots in the UAE.	Under discussion since 2025



An ADNOC-branded humanoid at ADIPEC 2026. Photo: Erik Arazas / Emarat Al Youm

We will track each of these in the Robotics MENA database, and report back when the numbers arrive.

07 CHAPTER 07

About the providers

Gecko Robotics

geckorobotics.com

Makes Raptor and the crawlers used across ADNOC Gas since 2023; signed a new ADNOC Onshore agreement at ADIPEC 2026.

US inspection-robot maker with a UAE base under NextGen FDI, deploying robots and AI across ADNOC operations.

In the fleet: Raptor

In the Robotics MENA directory ↗

Taurob

taurob.com

Built the explosion-proof inspector running at Taweelah and is a partner on the ARGOS operator robot.

Austrian (Vienna) maker of explosion-proof inspection robots; its Inspector is deployed at ADNOC's Taweelah gas plant in the UAE, and it partners on the ARGOS heavy-duty operator robot.

In the fleet: Taurob inspector, ARGOS

In the Robotics MENA directory ↗

Unitree Robotics

unitree.com

ADNOC Onshore agreed to test its robots, including humanoids, in field operations.

Chinese (Hangzhou) maker of quadruped and humanoid robots, listed in Shanghai; ADNOC Onshore signed at ADIPEC 2026 to test its robots in UAE field operations.

In the fleet: Unitree robots

In the Robotics MENA directory ↗

SeaOwl

seaowlgroup.com

Partner on ADNOC L&S's two 60-metre remotely operated landing craft.

French marine and energy services group (Paris) that pioneered remotely operated ships; developed ADNOC L&S's 60m remotely operated landing craft in the UAE, where it has a Ras Al Khaimah office.

In the fleet: Remote landing craft

In the Robotics MENA directory ↗



AIQ

aiq.ae

Joint venture of ADNOC and Presight (a G42 company). Part of the three-year inspection programme with Gecko and ADNOC Gas, taking Gecko's outputs across ADNOC.

Abu Dhabi AI-for-energy joint venture of ADNOC and G42's Presight; made the RoboWell autonomous well-control system and is in a three-year AI and robotics inspection programme with ADNOC Gas and Gecko.

In the Robotics MENA directory ↗

Sources

1 How wall-climbing robots could help Gulf energy producers recover from attacks · The National · 4 Nov 2026
https://www.thenationalnews.com/future/technology/2026/11/04/how-wall-climbing-robots-could-help-gulf-energy-producers-recover-from-attacks/

2 ADNOC Onshore and Gecko Robotics agreement signed at ADIPEC 2026 · ADNOC Group on X · 5 Nov 2026
https://x.com/ADNOCGroup/status/2118279593908142426

3 ADNOC signs agreement with OpenAI and G42 to deploy Frontier AI · The National · 2 Nov 2026
https://thenationalnews.com/business/energy/2026/11/02/adnoc-signs-agreement-with-openai-and-g42-to-deploy-frontier-ai

4 Adnoc expands AI across energy operations, with 70 robotic use cases · Khaleej Times · 3 Nov 2026
https://www.khaleejtimes.com/uae/adnoc-expansion-ai-robotics-energy-operations

5 ADNOC Onshore agreement with Unitree to test robotics and humanoid technologies (Arabic) · Attaqa (الطاقة) · 4 Nov 2026
https://attaqa.net/?p=651656

6 Robots on show at ADIPEC 2026 (Arabic) · Emarat Al Youm · 3 Nov 2026
https://www.emaratalyoum.com/business/local/2026-11-03-1.2098674

7 Adnoc's approach has made the UAE a 'top five' robotic country · The National · 25 Aug 2026
https://www.thenationalnews.com/future/technology/2026/08/25/gecko-robotics-uae-adnoc/

8 Gecko Robotics chief says partnership with Adnoc and UAE stronger than ever · The National · 27 Mar 2026
https://www.thenationalnews.com/future/technology/2026/03/27/gecko-robotics-jake-loosararian-adnoc/

9 US Navy IDIQ expansion · Gecko Robotics · 20 Oct 2026
https://www.geckorobotics.com/news/navy-idiq-expansion

10 ADNOC Gas, AIQ and Gecko Robotics programme · ADNOC Gas · 2 Nov 2025
https://adnocgas.ae/en/news-and-media/press-releases/2025/gecko-aiq

11 Gecko to help ADNOC Gas revolutionize asset operations and maintenance · Gecko Robotics · 29 Apr 2024
https://www.geckorobotics.com/news/gecko-to-help-adnoc-gas-revolutionize-asset-operations-and-maintenance

12 ADNOC deploys heavy-duty inspector robot for hazardous environments · Hydrocarbon Processing · 21 May 2026
https://hydrocarbonprocessing.com/news/2026/05/adnoc-deploys-heavy-duty-inspector-robot-for-hazardous-environments/

13 Heavy-duty robot takes over hazardous inspections at UAE gas plant · Interesting Engineering · 22 May 2026
https://interestingengineering.com/ai-robotics/heavy-duty-robot-uae-gas-plant

14 ADNOC Touts Robotic Deployment With More to Come · JPT · 12 Jun 2026
https://jpt.spe.org/adnoc-touts-robotic-deployment-with-more-to-come

15 Walking Into the Future: ADNOC Drilling Unveils First AI-Powered Island Rig · OE Digital · 25 Jun 2026
https://www.oedigital.com/news/540638-walking-into-the-future-adnoc-drilling-unveils-first-ai-powered-island-rig

16 ADNOC L&S unveils the UAE's first autonomous offshore vessels · Zawya (ADNOC L&S release) · 13 Nov 2025
https://www.zawya.com/en/press-release/companies-news/adnoc-ls-unveils-the-uaes-first-autonomous-offshore-vessels-to-enhance-operational-efficiency-and-increase-sustainability-503125

17 ADNOC L&S Presents UAE's First Autonomous Offshore Vessels · OE Digital · 14 Nov 2025
https://www.oedigital.com/news/532459-adnoc-l-s-presents-uae-s-first-autonomous-offshore-vessels

18 ADNOC targets 5m barrels per day with AI fleet efficiency gains · Riviera · 18 Dec 2025
https://www.rivieramm.com/news-content-hub/adnoc-targets-5m-barrels-per-day-with-ai-fleet-efficiency-gains-87210

19 Here's a list of Gulf energy infrastructure damaged in Iran war · gCaptain / Bloomberg · 12 Apr 2026
https://gcaptain.com/heres-a-list-of-gulf-energy-infrastructure-damaged-in-iran-war/

20 Adnoc Gas to restore 80% of Habshan complex by 2026-end after attack damage · Khaleej Times · 12 May 2026
https://www.khaleejtimes.com/business/adnoc-gas-says-it-will-restore-80-of-habshan-capacity-by-2026-end

Robotics
MENA

Robotics MENA is an independent, curated record of robotics, physical AI and automation across the Middle East and North Africa, for everyone who follows robotics in the region.

Get the weekly MENA robotics brief and explore the full database at

roboticsmena.com

contact@roboticsmena.com

Images are credited to their owners and used for identification and commentary.

© 2026 Robotics MENA · Ad-free and independent